

Nanyang Research Programme

CEE01 – Levelized cost of electricity for Pressure-Retarded Osmosis Plant

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Abstract

Pressure-retarded osmosis (PRO), a renewable energy harvesting process that harnesses osmotic energy from salinity gradients, offers a compelling alternative due to its controllability for power generation. To assess the feasibility of PRO plants for real-world applications, a comprehensive economic analysis is essential. This study investigates the levelized cost of electricity (LCOE) for a 1MW PRO plant as a function of membrane power density. A conventional counter-current flow PRO plant configuration is analysed, with river water as the feed solution and seawater as the draw solution. The total cost of the process is estimated by calculating both capital expenditure (CapEx) and operational expenditure (OpEx) using established equations from prior research. By analysing a range of water flux values and calculating the power density, a relationship between power density and LCOE was determined and visualized. Results indicated that a power density of 10 W/m² and a water flux of 25 l/m²/h resulted in an LCOE of S\$0.653/kWh. However, our findings showed that the LCOE of the PRO plant cannot compete with that of the electricity price and solar energy LCOE in Singapore with the lowest possible LCOE of the PRO plant approaching S\$0.559/kWh (when power density approaches infinity). These findings underscore the critical role of power density in determining the economic viability of PRO and establish benchmarks for future improvements in system design and membrane fabrication. Nevertheless, PRO's ability to provide continuous and controllable energy supply positions it as a promising renewable energy option.

Keywords: Pressure retarded osmosis, Levelized Cost of Electricity (LCOE), Economic analysis, Power density, Renewable energy

1 INTRODUCTION

With the rising issue of climate change and global warming, there has been increasing emphasis on renewable forms of energy. Due to research and cost reductions, energy sources like wind and solar energy have become competitive with fossil fuel energy. However, these forms of energy are dependent on external conditions, hence they may be unreliable and intermittent. Thus, it is necessary to develop more ways to generate renewable energy and lessen CO₂ emissions, especially one that is non-intermittent. One such technology that is promising is pressure-retarded osmosis (PRO). PRO harvests electricity by harnessing the osmotic energy from the salinity gradient between two solutions. The process is regulated by controlling the mixing of the solutions, allowing for a reliable and steady source of electricity harvesting as needed. Plenty of research has been conducted on PRO, in 1976, Loeb et al. published useful results on PRO [1]. More recently, membrane technology has had huge advances and more papers have analysed PRO plants. Currently, one of the main hurdles for PRO to be commercialised is cost-effectiveness, and whether the technology is developed sufficiently to make a profit. This makes it important to investigate the economic viability of such PRO plants to commercialise them. If the economic viability of PRO plants can be proven, it

becomes much more likely for it to become commercialised.

However, the scope of this study may be limited as the economic framework used is of conventional counter-current flow PRO. As the research of PRO gains popularity, many other modified forms of PRO are being discussed, which may be more efficient than conventional PRO. Despite this, studying the economic viability of conventional PRO can still be useful as it provides an economic baseline for the newer forms of PRO. This research paper aims to study what conditions are required for a PRO plant with a total power capacity of 1MW to be economically viable, and whether current PRO technology can be commercialized.

2 METHODOLOGY / MATERIALS

2.1 PRO system model conventional counter-current flow module

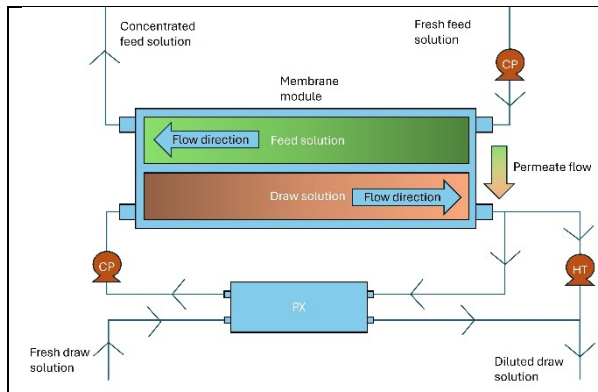


Figure 1. counter-current flow PRO module process

The conventional single stage PRO system used is an open loop, where both the draw solution and the feed solution are pressurized and continuously supplied into the membrane before it is discharged out of the system. The fresh draw solution is pressurized by exchanging hydraulic pressure with the diluted draw solution using a pressure exchanger (PX), and further pressurized by a circulation pump. The feed solution is pumped into the membrane module by a second circulation pump. Under a constant hydraulic pressure below the osmotic pressure difference ($\Delta\pi$), water molecules permeate through the membrane from the feed solution to the draw solution, which dilutes the draw solution. The diluted draw solution is then split into two streams. The first diluted draw solution stream that enters the PX will have the same flow rate as the original fresh draw solution. The other stream that passes through a hydro-turbine (HT) to generate electricity will have the same flow rate as the permeate flow rate [2].

2.2 Economic Analysis

The main metric used is the levelized cost of electricity (LCOE) which accounts for CapEx and OpEx. LCOE is a widely used metric to assess the financial feasibility of an electricity-generating product and is computed by taking average total cost of building and operating the power-generating asset over its lifetime divided by its total energy output. By comparing the LCOE of other forms of renewable forms of energy, the economic viability of PRO can be assessed.

In this study, it was assumed that total power capacity of the plant would be 1MW. It was further assumed that river water and seawater was used in the PRO system, so their salinity values of 0.6 M/l and 0.015 M/l respectively were used to calculate applied pressure. This was done using Van Hoff's Equation and assuming applied pressure to be half of osmotic pressure. The water flux was varied from values of 1 to 50 $\text{l/m}^2/\text{h}$. From this, the membrane power density was taken as

$$P_d = J_w \times \Delta P$$

where J_w is water flux and ΔP is the applied pressure, which is constant at 14.48 bar.

Subsequently, the membrane area needed was calculated by dividing the total power capacity with the power density. The membrane price ($\$10/\text{m}^2$) of the PRO membranes was assumed to be the same as the RO membranes.

Then, the permeate flow rate, Q_p was calculated from

$$Q_p = J_w \times A_m$$

where A_m is the membrane area.

Feed flow rate, Q_f , was taken as

$$Q_f = \frac{Q_p}{R}$$

where R is the PRO water recovery, assumed to be 0.5.

Finally, Q_d , draw flow rate, was calculated based on the ratio of Q_d to Q_f that is fed into the PRO module. All parameters for the PRO plant evaluated in this study are listed in Table 1.

Table 1: PRO LCOE parameters

Parameter	Symbol	Units	Value
Total Power Capacity	P	MW	1
PRO Water Recovery	R	-	0.5
Ratio of Q_d to Q_f	-	-	1
Draw Solution Concentration	w_d	M/l	0.6

Feed Solution Concentration	w_f	M/l	0.015
Applied Pressure	ΔP	bar	14.48
Interest Rate	i	%	5
Number of Periods of Loan	n	years	25
Membrane Cost	C_m	\$/m ²	10
Membrane Life	Y_m	years	5
Plant Life	PI	years	25
Exchange rate (euros to SGD)	ex	-	1.06
Capital Charge Rate	CC	%	8

PRO is more complex than RO because it involves two streams and two pumps. Therefore, for the intake and pretreatment calculations, both the feed flow rate of the feed and draw solutions were considered.

Lastly, it was assumed that there was minimal membrane fouling in the PRO system, thus the cost of maintenance from fouling is negligible. It was also further assumed that the cost of intake and pretreatment system conducted for RO would be a reasonable estimate for PRO. The post treatment, however, was not included as it is unnecessary in the PRO process.

The total cost of the PRO plant was broken down to two main groups: CapEx and OpEx. End of life costs were neglected in this paper.

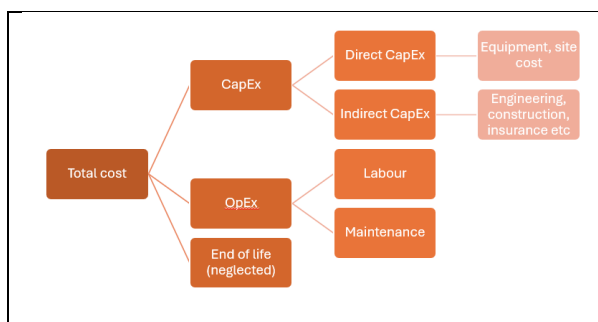


Figure 2. Breakdown of the costs of PRO plant

The CapEx of the main components of the PRO system that were included are the intake system ($CapEx_{intake}$), pretreatment ($CapEx_{pt}$), pressure exchanger ($CapEx_{P.Ex.}$), membrane module ($CapEx_m$), hydro turbine ($CapEx_{hyd.tur.}$) and pumps ($CapEx_{pumps}$). Below are the equations used [3-9]:

$$CapEx_{intake} = 598 \times (Q_d + Q_f) / 0.9^{0.78}$$

$$CapEx_{pt} = 697 \times 0.7 \times ((Q_d + Q_f) / 0.9)^{0.78}$$

$$CapEx_{P.Ex.} = 146 \times (Q_d \times 0.024)^{0.478}$$

$$CapEx_m = C_m \times A_m \times \frac{PI}{Y_m}$$

$$CapEx_{hyd.tur.} = (2600(P)^{0.58} \times ex)$$

$$CapEx_{pumps} = 52 \times (\Delta P \times Q_d)^{0.8}$$

The equipment cost was the sum of intake, pretreatment, pumps, pressure exchanger, hydro turbine and membrane.

$$CapEx_{equip} = CapEx_{intake} + CapEx_{pt} + CapEx_{P.Ex.} + CapEx_m + CapEx_{hyd.tur.} + CapEx_{pumps}$$

The site cost was estimated to be 0.2 times of the equipment cost. The direct cost was the sum of the site and equipment cost while indirect cost which included the cost of engineering, construction, insurance, etc, was 0.3 times the direct cost.

$$CapEx_{site} = CapEx_{equip} \times 0.2$$

$$Direct\ CapEx = CapEx_{equip} + CapEx_{site}$$

$$Indirect\ CapEx = Direct\ CapEx \times 0.3$$

$$Total\ CapEx = Indirect\ CapEx + Direct\ CapEx$$

To find the annual CapEx, the total CapEx was multiplied with the Capital Recovery Factor where the interest rate, i , was assumed to be 5% and the plant life, n , is 25 years.

$$Annual\ CapEx = Total\ CapEx \times \frac{i(1+i)^n}{(1+i)^n - 1}$$

The annual operating costs were then taken as

$$Annual\ OpEx = 1.136 \times CC \times Annual\ CapEx$$

Finally, the LCOE was calculated by

$$LCOE = \frac{Annual\ OpEx + Annual\ CapEx}{W_{net} \times t}$$

Where W_{net} is net power and t is the annual operation time.

3 RESULTS/DISCUSSION

3.1 Higher power density reduces LCOE

In this paper, the water flux was varied across a range of values to vary the power density and investigate the minimum power density needed to

achieve an ideal LCOE. With a higher water flux, power density increases linearly. Thus, membrane area decreases, leading to a lower membrane CapEx. From fig. 3, it can be seen that CapEx_m has an inverse relationship with power density and water flux. Since water flux is proportional to power density, each water flux value corresponds to a specific power density and the graphs overlap.

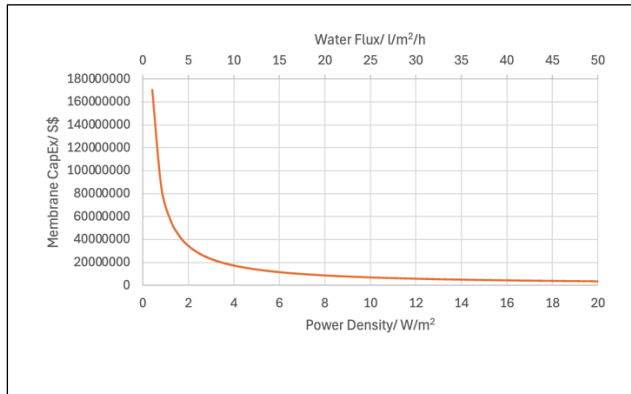


Fig. 3. CapEx_m as a function of Power Density and Water Flux with an inverse relationship.

From Fig. 4, since water flux increases and power density increases, membrane area needed to achieve a 1MW PRO plant decreases. As all other parameters remain the same, the annual CapEx decreases and thus, LCOE decreases. Similar to the Membrane Cost graphs, the graphs of LCOE vs Water Flux and LCOE vs Power Density overlap. Particularly, for this PRO plant, a power density of 10 W/m^2 and a water flux of 25 $\text{l/m}^2/\text{h}$ resulted in an LCOE of S\$0.653/kWh. For a water flux of 50 $\text{l/m}^2/\text{h}$, the power density is 20 W/m^2 , and the LCOE can be reduced to S\$0.606/kWh.

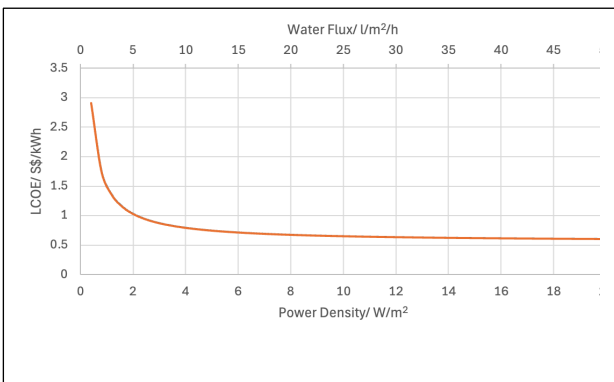


Fig. 4. LCOE as a function of power density and water flux.

The LCOE of the PRO plant was compared to that of the electricity and renewable energy in Singapore to determine economic viability. The cost of fuel and power generation of electricity in Singapore is S\$0.214/kWh. The fuel cost is the cost of imported natural gas, which is tied to oil prices by commercial contracts. The cost of power

generation covers mainly the costs of operating the power stations, such as the manpower and maintenance costs, as well as the capital cost of the stations [11].

As for renewable energy, the LCOE of the PRO plant was compared to that of solar photovoltaic (PV). Singapore's high average annual solar irradiation of about 1,580 kWh/m^2 makes PV a potential renewable energy option for Singapore. Singapore aims to increase solar capacity to at least 2 gigawatt-peak by 2030, equivalent to powering about 350,000 households a year. According to Energy Market Authority (EMA), using solar energy at home ranges from S\$0.11/kWh to S\$0.15/kWh in Singapore based on estimates [10]. While solar energy is a popular renewable energy source, a significant advantage of PRO compared to solar energy is that it is non intermittent. It can be conducted despite of weather conditions whereas PV can only be collected in sunny conditions, which means that it energy can only be generated for 8 hours in a day assuming that the weather is ideal.

However, as $J_{wi} \rightarrow \infty$ and $P_d \rightarrow \infty$, $\text{LCOE} \rightarrow \text{S\$0.559/kWh}$. Therefore, it is impossible for the 1MW PRO plant to achieve a LCOE similar to solar and fossil fuel energy.

Thus, PRO plants must not only reduce pretreatment and intake costs but also achieve higher power densities than those shown to be achievable on module scale PRO plants conducted in other research papers to be competitive with solar and fossil fuel energy.

In 2013, the highest power density achieved was by Straub et. al, with a power density of up to 60 W/m^2 with a 3 M (180 g/L) NaCl draw solution and a 7.7cm wide and 2.6cm long membrane area [12]. However, it was mentioned by Hyung, et. Al, that even though the module power density is high, this system likely had negative net power density. As the system size is enlarged, module power density decreases because the average driving force decreases [13].

In 2018, Kurihara, et. Al established a PRO hollow fibre membrane module with a maximum 13.3 W/m^2 of membrane power density using a 10-inch module at a prototype PRO plant as part of the "Megaton Water System" project. They also reported power density reduction from 17.1 to 13.3 W/m^2 as the module length decreased from 10 in to 5 in. Thus, it is uncertain whether a PRO plant of such a large size is able to achieve as high of a power density [14].

For PRO to be economically viable and compete with other forms of electricity generation commercially, it must be able to achieve higher power densities during operation, which may need to be even higher than what is stated in this paper. Ways to increase the power density of PRO plants are discussed below, namely by investigating ways to improve commercial PRO membranes and enhance the conventional PRO process.

3.2 Development of PRO membranes

PRO membranes are another issue when it comes to commercialising PRO plants. Ideally, PRO membranes should have strong mechanical strength to withstand relatively high pressures, which is optimally half of the osmotic pressure, have high salt rejection, high water permeability to enhance water flux to achieve high power density, and low structure parameter to minimise internal concentration polarisation (ICP) [15]. Additionally, membranes should also have anti-fouling potential, though many pretreatment approaches to suppress fouling have been studied [16].

However, it is difficult to find a membrane that fulfils all these criteria. Many commercial membranes do not possess the characteristics of both forward osmosis membranes to reduce concentration polarisation, and reverse osmosis membranes to withstand high hydraulic pressures [17]. Current commercially available membrane modules and spacers used in PRO also can cause channel blockage and severe membrane deformation, increasing pressure drop [18].

These challenges that commercialised membranes face reduce the overall performance of PRO, reducing the economic viability of having a commercialised plant. This issue hence has become a topic of interest to several researchers, leading to new innovations for PRO membranes. Rather than using common cellulose acetate (CA), cellulose triacetate (CTA) or thin-film composite (TFC) membranes [19] new materials for PRO membranes are currently being researched.

Zhang et al. developed a thin Polyethersulfone (PES) composite hollow fibre membrane, achieving a power density of 24.3 W/m² when 1 M NaCl and DI water were evaluated at 0.2 MPa for FO operation [20].

Fan et al. also worked on the construction of reduced graphene oxide on carbon nanotubes (CNT) hollow fibre substrates through electrophoretic deposition and a chemical-reduction process for membranes on nanocarbon substrates. The construction of a graphene active

layer has shown to improve permeability and ion selectivity [21].

While many of these membranes are only lab-scale, with further research into PRO membranes, an inexpensive, commercialised membrane with high mechanical strength, high selectivity and high water permeability may be formed, which would greatly increase the likelihood of commercialising PRO.

3.3 Optimising PRO

Based on the power density calculation formula presented in Section 2.2, another approach to enhancing power density is to increase the pressure applied in the PRO process. However, since conventional PRO plants operate under a constant-pressure mode, there is limited operating pressure, resulting in under-pressurization (UP) and thus lowering the energy production. Energy conversion efficiency is low due to incomplete utilization of the potential osmotic pressure difference between feed and draw solutions. In addition, there is often high internal concentration polarisation on the membrane which reduces driving force.

There have been efforts to find different modes of PRO that can reduce these issues. One such common one is multi-stage PRO (MSPRO) [15]. Due to the many stages of PRO, hydraulic pressure can be closer to the osmotic pressure difference to achieve a higher energy efficiency. Furthermore, it taps on much of the remaining potential energy in the feed and draw solution of PRO as the single stage system only passes the water through one membrane module without staged optimisation. This is a viable method for increasing energy production, however, cost largely increases due to increase in stages.

Another solution proposed was the atmospheric batch PRO (ABPRO) [2] where an atmospheric tank is used to receive and store intermediate diluted draw solution and PX to recover pressure energy from diluted draw solution before being recycled. Compared to the conventional single-stage PRO (SSPRO), ABPRO can approach thermodynamic maximum energy production under ideal conditions. The energy loss caused by the UP effect can be mitigated in ABPRO due to the lower per-cycle water recovery and the variable-pressure mode. However, it induces the energy loss owing to entropy generation.

To address the exponentially increasing capital costs associated with additional stages in MSPRO and minimize entropy generation caused by mixing

in ABPRO, the semi-closed PRO (SCPRO) configuration has been proposed [2]. This design incorporates two atmospheric tanks to enable multi-cycle operation, facilitating the separate storage of intermediate fresh draw solution and diluted draw solution in each cycle. By mitigating energy losses due to the UP effect without introducing additional issues, SCPRO achieves a promising energy efficiency of approximately 75%.

Furthermore, the low salinity gradient between commonly used draw solutions (e.g., seawater) and feed solutions (e.g., river water) imposes a limitation on the applied hydraulic pressure, thereby restricting the achievable power density of PRO plants.

Various pairs of draw and feed solutions with a greater concentration difference, such as Dead Sea water and river water, have been utilized in PRO plants for osmotic energy harvesting, indicating a promising performance. However, this approach significantly restricts the feasible locations for PRO plant deployment. Therefore, the Desalination-Osmotic Energy Storage (DOES) System [23] has been put forward. This system utilizes the brine and partial of the permeate produced by reverse osmosis (RO) as the draw solution and feed solution, respectively, in PRO for power generation. Internal fouling is mitigated due to use of clean desalinated freshwater as feed solution which increases the osmotic power harvesting. The higher salinity gradient between feed and draw solution with a larger RO recovery also increases power density. This system possibly mitigates the mismatch of demand and supply in the electricity and water sectors while also reducing LCOE of the plant.

5 CONCLUSION

By calculating the total cost of a PRO plant and assigning a series of power density values, the corresponding values of LCOE was calculated. A graph of LCOE against power density was then plotted, where it was found that power density has an inverse relationship with LCOE.

It was calculated that a power density of 10 W/m² and a water flux of 25 l/m²/h resulted in an LCOE of S\$0.653/kWh. However, the LCOE of PRO is unable to match Singapore's electricity price of S\$0.2136/kWh, the lowest possible LCOE that the PRO plant approaches is S\$0.559/kWh (when power density approaches infinity).

Moreover, while high power density values have been achieved before in module-scale PRO plants

by previous researchers, it may not be currently feasible in a commercial large scale PRO plant, as power density was observed to decrease as the PRO module scale was increased. As lower bound cost estimate was used and due to the assumptions made in this paper, the cost of a large-scale PRO plant is also likely to be higher than what was found, indicating that the LCOE of a large scale PRO plant may be even higher.

Although the LCOE of PRO plants using seawater as the draw solution and river water as the feed solution is challenging to compete with that of solar energy, the ease of operation and control still makes PRO plants a promising alternative.

Looking ahead, further reducing the LCOE of PRO and enhancing its economic viability will require optimizing PRO process designs and operating modes, utilizing solutions with higher salinity gradients, and advancing the performance of commercial PRO membranes.

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